Figure 2.4 Data lake socio-technical architecture

A data lake is often described as more than a storage pattern. It also offers many related capabilities (ingestion mechanisms, security, metadata, catalog, processing engines, access mechanisms, and many others).

The common socio-technical architecture of the data lake is almost the same as the enterprise data warehouse. As shown in figure 2.4, all data-related components are owned by the central data team. Here's an example of when you should consider implementing a data lake:

- You deal with big data (volume, velocity, and variety).
- You have structured, semi-structured, and unstructured sources of data.
- It is hard to foresee all your data use cases.
- Data sources are owned by a small number of teams.
- Consumption patterns are not yet established.

2.2.3 Data lakehouse

A *data lakehouse* is a combination of a data warehouse and data lake (figure 2.5). It is built on top of data lake with added features like these:

- Atomicity, consistency, isolation, and durability (ACID) transaction support
- Schema enforcement and governance
- BI support on top of the same data source